MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, 90-DEGREE H-PLANE BEND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.

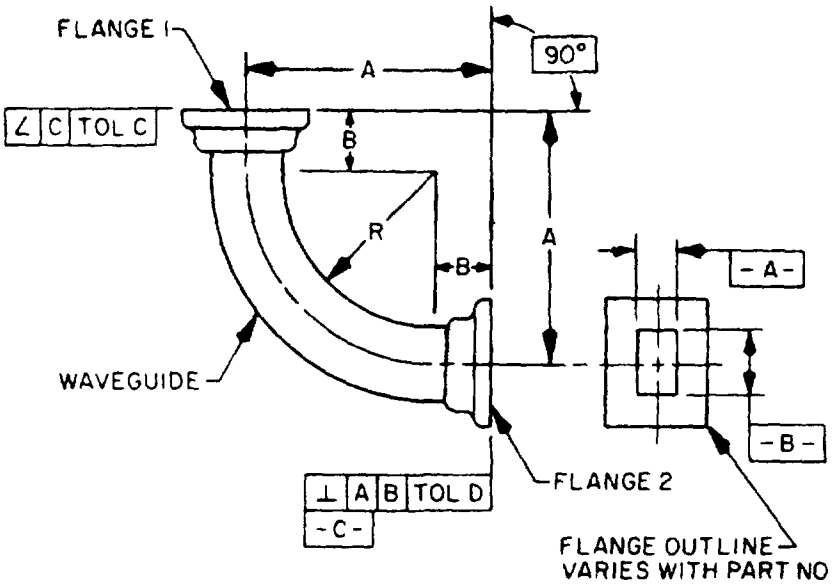


FIGURE 1 Assembly configuration

FSC 5985

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TABLE I. Dash number, electrical and physical characteristics, and dimensions.

Dash No.	Waveguide M85/	Flange 1 M3972/	Flange 2 M3972/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lbf/in^2 gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
001	1-043	56-001	61-002	2.6 - 3.95	-	1.05	15	10.500 (266.70) ±.030 (.76)	1.500 (38.10)	.115 (2.92)	.012 (.30)	8.250 (209.55) ±.125 (3.18)
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	30	7.000 (177.80) ±.030 (.76)	1.000 (25.40)	.077 (1.96)	.008 (.20)	5.500 (139.70) ±.125 (3.18)
003	1-067	55-001	60-001	5.85 - 8.20	-	1.03	30	4.875 (123.83) ±.025 (.64)	.750 (19.05)	.056 (1.42)	.006 (.15)	3.750 (95.25) ±.125 (3.18)
004	1-073	53-002	59-007	7.05 - 10.0	-	1.07	30	4.062 (103.17) ±.020 (.51)	.687 (17.45)	.044 (1.12)	.005 (.13)	3.063 (77.80) ±.063 (1.60)
005	1-079	53-001	59-006	8.20 - 12.4	-	1.05	30	3.562 (90.47) ±.020 (.51)	.562 (14.27)	.038 (.97)	.004 (.10)	2.750 (69.85) ±.063 (1.60)
006	1-102	54-001	59-003	18.0 - 26.5	.11	1.05	30	1.722 (43.74) ±.015 (.38)	.410 (10.41)	.019 (.48)	.002 (.05)	1.188 (30.18) ±.063 (1.60)
007	1-106	54-001	59-003	18.0 - 26.5	.11	1.05	30	1.722 (43.74) ±.015 (.38)	.410 (10.41)	.019 (.48)	.002 (.05)	1.188 (30.18) ±.063 (1.60)
008	1-077	53-003	59-008	8.2 - 12.4	-	1.05	30	3.562 (90.47) ±.015 (.38)	.562 (14.27)	.038 (.97)	.004 (.10)	2.750 (69.85) ±.063 (1.60)
009	1-071	53-004	59-009	7.05 - 10.0	-	1.07	30	4.062 (103.17) ±.020 (.51)	.687 (17.45)	.044 (1.12)	.005 (.13)	3.063 (77.80) ±.063 (1.60)
010	1-041	56-002	61-001	2.60 - 3.95	-	1.05	15	10.500 (266.70) ±.030 (.76)	1.500 (38.10)	.115 (2.92)	.012 (.30)	8.250 (209.55) ±.125 (3.18)

See footnotes at end of table.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued.

Dash No.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lb_f/in^2	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.04	30	2.601 (66.07) ±.015 (.38)	.500 (12.70)	.028 (.71)	.003 (.08)	1.914 (48.62) ±.063 (1.60)
012	1-053	57-001	62-001	3.95 - 5.85	-	1.05	30	7.000 (177.80) ±.030 (.76)	1.000 (25.40)	.076 (1.93)	.008 (.20)	5.500 (139.70) ±.125 (3.18)
013	3-006	54-003	59-005	26.5 - 40.0	.15	1.05	30	1.242 (31.55) ±.010 (.25)	.312 (7.92)	.014 (.36)	.001 (.03)	.828 (21.03) ±.063 (1.60)
014	1-065	55-002	60-002	5.85 - 8.20	-	1.03	30	4.875 (123.82) ±.025 (.64)	.750 (19.05)	.056 (1.42)	.006 (.15)	3.750 (95.25) ±.125 (3.18)
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.04	30	2.601 (66.07) ±.015 (.38)	.500 (12.70)	.028 (.71)	.003 (.08)	1.914 (48.62) ±.063 (1.60)
016	1-103	54-002	59-004	18.0 - 26.5	.11	1.05	30	1.722 (43.74) ±.015 (.38)	.410 (10.41)	.019 (.48)	.002 (.05)	1.188 (30.18) ±.063 (1.60)
017	1-018	52-002	52-002	1.12 - 1.70	-	1.05	30	7.810 (198.37) ±.010 (.25)	1.116 (28.35)	.106 (2.69)	.028 (.71)	4.989 (126.72) ±.125 (3.18)
018	3-006	54-003	54-003	26 5 - 40	.15	1.1	30	1.000 (25.40) ±.015 (.38)	.500 (12.70)	.013 (.33)	.001 (.03)	.500 (12.70) ±.015 (.38)
019	3-006	67-005	67-005	26.5 - 40	.15	1 1	---	↓	↓	.014 (.36)	↓	↓
020	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	---	↓	↓	↓	↓	↓
021	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	---	↓	↓	↓	↓	↓

See footnotes at end of table.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued.

Dash No.	Waveguide MBS/	Flange 1	Flange 2	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lb_f/in^2 gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
		M3922/		GHz	dB							
022	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	---	1.000 (25.40) ±.015 (.38)	.500 (12.70)	.014 (.36)	.001 (.03)	.500 (12.70) ±.015 (.38)
023	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	---	1.000 (25.40) ±.015 (.38)	.500 (12.70)	.012 (.30)	.001 (.03)	.500 (12.70) ±.015 (.38)
024	1-001	76-07	76-07	.32 - .49	.10	1.05	.25	51.000 (1295.40) ±.015 (.38)	3.000 (76.20)	.561 (14.25)	.102 (2.59)	36.312 (922.32) ±.015 (.38)
025	1-003	76-06	76-06	.35 - 53	.10	1.05	25	45.000 (1143.00) ±.015 (.38)	3.000 (96.20)	.500 (12.70)	.093 (2.36)	31.312 (729.32) ±.015 (.38)
026	1-005	76-05	76-05	.410 - .625	.10	1.05	.25	39.000 (990.60) ±.015 (.38)	3.000 (76.20)	.434 (11.02)	.080 (2.03)	26.875 (682.62) ±.015 (.38)
027	1-007	76-04	76-04	.490 - 750	.10	1.05	.25	33.000 (838.20) + 015 (.38)	3.000 (76.20)	.369 (9 37)	.067 (1.70)	22.375 (568.32) +.015 (.38)
028	1-009	76-03	76-03	.64 - 96	.10	1.05	50	27.000 (685.80) ±.015 (.38)	3.000 (76.20)	.301 (7.65)	.051 (1 30)	18.125 (460.38) ±.015 (.38)
029	1-011	76-02	76-02	.75 - 1 12	10	1 05	50	21.000 (533.40) ±.015 (.38)	3.000 (76.20)	.241 (6.12)	.044 (1.12)	13.000 (330.20) ±.015 (.38)
030	1-013	76-01	76-01	96 - 1 45	.10	1.05	1.0	14.000 (355.60) ±.015 (.38)	2.000 (50.80)	.171 (4.34)	.035 (.89)	8.025 (203.84) ±.015 (.38)
031	1-018	52-002	52-002	1 12 - 1.70	10	1.05	1.5	14.000 (355.60) ±.015 (.38)	2.000 (50.80)	.160 (4.06)	.029 (.74)	8.670 (220.22) ±.015 (.38)
032	1-029	52-006	52-006	1.70 - 2 60	.10	1.05	3.0	10.000 (254.00) ±.015 (38)	2.000 (50.80)	.101 (2.57)	.019 (.48)	5.770 (146.56) +.015 (.38)

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.

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REQUIREMENTS

Dimensions and configuration: See figure 1 and table I.

Electrical characteristics: See table I.

Part number: M3970/13-(dash number from table I).

Cross reference. See table II.

TABLE II. Cross reference.

Part number M3970/13-	Type designation	Part number M3970/12-	Type designation
001	UG-712/U	009	UG-720/U
002	UG-713/U	010	UG-721/U
003	UG-714/U	011	UG-722/U
004	UG-715/U	012	UG-723/U
005	UG-716/U	013	UG-724/U
006	UG-717/U	014	UG-725/U
007	UG-718/U	015	UG-726/U
008	UG-719/U	016	UG-727/U

NOTE Assemblies covered by this specification are substitutable for the following commercial types

Part number M3970/11	Manufacturer's code	Commercial type
017	10001	2938971-1
017	01456	650BH49
017	05865	L2257
017	17250	L11114
018	73293	44321H-902
019	73293	44322H-902
020	73293	44323H-902
021	73293	44324H-902
022	73293	44325H-902
023	73293	44326H-902
024	52564	3005001
025	52564	3105001
026	52564	3205001
027	52564	3305001
028	52564	3405001
029	52564	3505001
030	52564	3605001
031	52564	3705001
032	52564	4005001

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Navy - EC
Air Force - 85

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Navy - EC, OS
Air Force - 11, 17, 99
DSA - ES

User activities
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Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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